Newman-Penrose Formalism Calculations

Stephani et al., Equation (13.84_1)

Metric

$$ds^2 = -\mathrm{d}t \otimes \mathrm{d}t - 2\,ay\mathrm{d}t \otimes \mathrm{d}z - 2\,ay\mathrm{d}z \otimes \mathrm{d}t + \frac{4\left(\left(a^2+b^2\right)F\left(x\right)^2\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2 - \Lambda a^2y^2\right)}{\Lambda}\mathrm{d}z \otimes \mathrm{d}z + \mathrm{d}x \otimes \mathrm{d}x + \frac{4\left(a^2+b^2\right)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2}{\Lambda F\left(x\right)^2}\mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$(x^\mu) = (t, z, x, y)$

Metric Functions

$$Y = \frac{\sqrt{a^2+b^2}\sin\left(2\sqrt{\Lambda}x\right)}{\sqrt{\Lambda}}$$

$$F = e^{\left(-\frac{\sqrt{\Lambda}b}{2\left(a^2+b^2\right)\tan\left(2\sqrt{\Lambda}x\right)}\right)}$$

$$c = \sqrt{a^2+b^2}$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}\right)dt + \left(\sqrt{2}ay - \frac{\sqrt{2}\sqrt{a^2+b^2}F\left(x\right)\sin\left(2\sqrt{\Lambda}x\right)}{2\sqrt{\Lambda}}\right)dz$$

$$n = \left(\frac{1}{2}\sqrt{2}\right) dt + \left(\sqrt{2}ay + \frac{\sqrt{2}\sqrt{a^2+b^2}F(x)\sin\left(2\sqrt{\Lambda}x\right)}{2\sqrt{\Lambda}}\right) dz$$

$$m = \left(\frac{1}{2}\sqrt{2}\right) dx + \left(\frac{i\sqrt{2}\sqrt{a^2+b^2}\sin\left(2\sqrt{\Lambda}x\right)}{2\sqrt{\Lambda}F(x)}\right) dy$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}\right) dx + \left(-\frac{i\sqrt{2}\sqrt{a^2+b^2}\sin\left(2\sqrt{\Lambda}x\right)}{2\sqrt{\Lambda}F(x)}\right) dy$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}a^2}{2(a^2+b^2)} - \frac{\sqrt{2}b^2}{2(a^2+b^2)} + \frac{\sqrt{2}\sqrt{\Lambda}ay}{2\sqrt{a^2+b^2}F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\right) dt + \left(-\frac{\sqrt{2}\sqrt{\Lambda}}{4\sqrt{a^2+b^2}F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\right) dz$$

$$n = \left(-\frac{\sqrt{2}a^2}{2(a^2+b^2)} - \frac{\sqrt{2}b^2}{2(a^2+b^2)} - \frac{\sqrt{2}\sqrt{\Lambda}ay}{2\sqrt{a^2+b^2}F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\right) dt + \left(\frac{\sqrt{2}\sqrt{\Lambda}}{4\sqrt{a^2+b^2}F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\right) dz$$

$$m = \left(\frac{1}{2}\sqrt{2}\right) dx + \left(\frac{i\sqrt{2}\sqrt{\Lambda}F(x)}{4\sqrt{a^2+b^2}\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\right) dy$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}\right) dx + \left(-\frac{i\sqrt{2}\sqrt{\Lambda}F(x)}{4\sqrt{a^2+b^2}\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\right) dy$$

Directional Derivatives

$$D = -\frac{\sqrt{2}a^2}{2\left(a^2+b^2\right)} - \frac{\sqrt{2}b^2}{2\left(a^2+b^2\right)} + \frac{\sqrt{2}\sqrt{\Lambda}ay}{2\sqrt{a^2+b^2}F\left(x\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\partial_t - \frac{\sqrt{2}\sqrt{\Lambda}}{4\sqrt{a^2+b^2}F\left(x\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\partial_z$$

$$\Delta = -\frac{\sqrt{2}a^2}{2\left(a^2+b^2\right)} - \frac{\sqrt{2}b^2}{2\left(a^2+b^2\right)} - \frac{\sqrt{2}\sqrt{\Lambda}ay}{2\sqrt{a^2+b^2}F\left(x\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\partial_t + \frac{\sqrt{2}\sqrt{\Lambda}}{4\sqrt{a^2+b^2}F\left(x\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\partial_z$$

$$\delta = \frac{1}{2}\sqrt{2}\,\partial_x + \frac{i\sqrt{2}\sqrt{\Lambda}F\left(x\right)}{4\sqrt{a^2+b^2}\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\partial_y$$

$$\bar{\delta} = \frac{1}{2}\sqrt{2}\,\partial_x - \frac{i\sqrt{2}\sqrt{\Lambda}F\left(x\right)}{4\sqrt{a^2+b^2}\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}\partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = \frac{\sqrt{\Lambda}a^2\cos\left(\sqrt{\Lambda}x\right)}{\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{\sqrt{\Lambda}b^2\cos\left(\sqrt{\Lambda}x\right)}{\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{a^2\frac{\partial}{\partial x}F\left(x\right)}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)F\left(x\right)} + \frac{b^2\frac{\partial}{\partial x}F\left(x\right)}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)F\left(x\right)}$$
$$- \frac{\sqrt{\Lambda}a^2}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} - \frac{\sqrt{\Lambda}b^2}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} - \frac{i\,\Lambda a}{4\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2}$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{\Lambda}a^2 \sin\left(\sqrt{\Lambda}x\right)}{\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)} + \frac{\sqrt{\Lambda}b^2 \sin\left(\sqrt{\Lambda}x\right)}{\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)} - \frac{a^2 \frac{\partial}{\partial x}F(x)}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)F(x)} - \frac{b^2 \frac{\partial}{\partial x}F(x)}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)F(x)}$$
$$- \frac{\sqrt{\Lambda}a^2}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} - \frac{\sqrt{\Lambda}b^2}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{i\,\Lambda a}{4\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2}$$

$$\alpha = -\frac{\sqrt{\Lambda}a^2 \cos\left(\sqrt{\Lambda}x\right)}{\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\sin\left(\sqrt{\Lambda}x\right)} - \frac{\sqrt{\Lambda}b^2 \cos\left(\sqrt{\Lambda}x\right)}{\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{a^2 \frac{\partial}{\partial x}F(x)}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)F(x)} + \frac{b^2 \frac{\partial}{\partial x}F(x)}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)F(x)}$$
$$+ \frac{\sqrt{\Lambda}a^2}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{\sqrt{\Lambda}b^2}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{i\,\Lambda a}{8\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2}$$

$$\beta = \frac{\sqrt{\Lambda}a^2 \cos\left(\sqrt{\Lambda}x\right)}{\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{\sqrt{\Lambda}b^2 \cos\left(\sqrt{\Lambda}x\right)}{\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\sin\left(\sqrt{\Lambda}x\right)} - \frac{a^2 \frac{\partial}{\partial x}F(x)}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)F(x)} - \frac{b^2 \frac{\partial}{\partial x}F(x)}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)F(x)}$$
$$- \frac{\sqrt{\Lambda}a^2}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} - \frac{\sqrt{\Lambda}b^2}{2\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} - \frac{i\,\Lambda a}{8\left(\sqrt{2}a^2+\sqrt{2}b^2\right)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2}$$

$$\tau = \frac{\sqrt{2}\sqrt{\Lambda}\sin\left(\sqrt{\Lambda}x\right)}{2\,\cos\left(\sqrt{\Lambda}x\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x}F(x)}{4\,F(x)} - \frac{\sqrt{2}\sqrt{\Lambda}}{4\,\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}$$

$$\pi = \frac{\sqrt{2}\sqrt{\Lambda}\cos\left(\sqrt{\Lambda}x\right)}{2\,\sin\left(\sqrt{\Lambda}x\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x}F(x)}{4\,F(x)} - \frac{\sqrt{2}\sqrt{\Lambda}}{4\,\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{2\,\Lambda a^4 \cos\left(\sqrt{\Lambda}x\right)^2}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{4\,\Lambda a^2b^2 \cos\left(\sqrt{\Lambda}x\right)^2}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{2\,\Lambda b^4 \cos\left(\sqrt{\Lambda}x\right)^2}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{\sqrt{\Lambda}a^4 \cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)}$$
$$- \frac{2\sqrt{\Lambda}a^2b^2 \cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)} - \frac{\sqrt{\Lambda}b^4 \cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)} - \frac{a^4 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{a^2b^2 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2}$$
$$- \frac{b^4 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{a^4 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{a^2b^2 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{b^4 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2}$$
$$+ \frac{2\,\Lambda a^4}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{4\,\Lambda a^2b^2}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{2\,\Lambda b^4}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{\sqrt{\Lambda}a^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)}$$
$$+ \frac{\sqrt{\Lambda}a^2b^2\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{\sqrt{\Lambda}b^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{a^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2}$$
$$+ \frac{a^2b^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{b^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{a^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2}$$
$$- \frac{a^2b^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{b^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2}$$
$$- \frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{\Lambda^2a^2}{16\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^4}$$

$$\Phi_{01} = 0$$

$$
\begin{aligned}
\Phi_{02} = {} & \frac{\sqrt{\Lambda}a^4\cos\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} + \frac{2\,\sqrt{\Lambda}a^2b^2\cos\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} + \frac{\sqrt{\Lambda}b^4\cos\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} - \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{3\,\sqrt{\Lambda}a^4\cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} - \frac{3\,\sqrt{\Lambda}a^2b^2\cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} \\
& - \frac{3\,\sqrt{\Lambda}b^4\cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} + \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} \\
& + \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{\sqrt{\Lambda}a^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)^3} \\
& + \frac{\sqrt{\Lambda}a^2b^2\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)^3} + \frac{\sqrt{\Lambda}b^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)^3} - \frac{a^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{a^2b^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{b^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{a^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{a^2b^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{b^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^4} \\
& + \frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{\Lambda^2a^2}{32\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^4}
\end{aligned}
$$

$$\Phi_{10} = 0$$

$$
\begin{aligned}
\Phi_{11} = {} & \frac{\Lambda a^4}{a^4+2\,a^2b^2+b^4} + \frac{2\,\Lambda a^2b^2}{a^4+2\,a^2b^2+b^4} + \frac{\Lambda b^4}{a^4+2\,a^2b^2+b^4} + \frac{\sqrt{\Lambda}a^4\cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)} + \frac{2\,\sqrt{\Lambda}a^2b^2\cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)} + \frac{\sqrt{\Lambda}b^4\cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)} \\
& - \frac{a^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2} - \frac{a^2b^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2} - \frac{b^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2} + \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} \\
& + \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} \\
& - \frac{\sqrt{\Lambda}a^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} - \frac{\sqrt{\Lambda}a^2b^2\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} - \frac{\sqrt{\Lambda}b^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} \\
& - \frac{a^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{a^2b^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{b^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{a^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} \\
& + \frac{a^2b^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{b^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{\Lambda^2a^2}{64\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^4}
\end{aligned}
$$

$$\Phi_{12} = 0$$

$$
\begin{aligned}
\Phi_{20} = {} & \frac{\sqrt{\Lambda}a^4\cos\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} + \frac{2\,\sqrt{\Lambda}a^2b^2\cos\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} + \frac{\sqrt{\Lambda}b^4\cos\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} - \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{3\,\sqrt{\Lambda}a^4\cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} - \frac{3\,\sqrt{\Lambda}a^2b^2\cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} \\
& - \frac{3\,\sqrt{\Lambda}b^4\cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^3} + \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} \\
& + \frac{a^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{a^2b^2\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{b^4\cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{\sqrt{\Lambda}a^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)^3} \\
& + \frac{\sqrt{\Lambda}a^2b^2\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)^3} + \frac{\sqrt{\Lambda}b^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)^3} - \frac{a^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{a^2b^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{b^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{a^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} \\
& - \frac{a^2b^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{b^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^4} \\
& + \frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^4} + \frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^4} - \frac{\Lambda^2a^2}{32\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^4}
\end{aligned}
$$

$$\Phi_{21} = 0$$

$$
\begin{aligned}
\Phi_{22} = & -\frac{2\,\Lambda a^4 \cos\left(\sqrt{\Lambda}x\right)^2}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{4\,\Lambda a^2b^2 \cos\left(\sqrt{\Lambda}x\right)^2}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{2\,\Lambda b^4 \cos\left(\sqrt{\Lambda}x\right)^2}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{\sqrt{\Lambda}a^4 \cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)} \\
& - \frac{2\,\sqrt{\Lambda}a^2b^2 \cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)} - \frac{\sqrt{\Lambda}b^4 \cos\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)} - \frac{a^4 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{a^2b^2 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} \\
& - \frac{b^4 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{a^4 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{a^2b^2 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{b^4 \cos\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} \\
& + \frac{2\,\Lambda a^4}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{4\,\Lambda a^2b^2}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{2\,\Lambda b^4}{(a^4+2\,a^2b^2+b^4)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{\sqrt{\Lambda}a^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} \\
& + \frac{\sqrt{\Lambda}a^2b^2\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{\sqrt{\Lambda}b^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)\sin\left(\sqrt{\Lambda}x\right)} + \frac{a^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} \\
& + \frac{a^2b^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{b^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{a^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} \\
& - \frac{a^2b^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{b^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2} \\
& - \frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{\Lambda^2a^2}{16\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^4}
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {} & \frac{\Lambda a^4}{3\left(a^4+2\,a^2b^2+b^4\right)} + \frac{2\,\Lambda a^2b^2}{3\left(a^4+2\,a^2b^2+b^4\right)} + \frac{\Lambda b^4}{3\left(a^4+2\,a^2b^2+b^4\right)} - \frac{2\,\Lambda a^4\cos\left(\sqrt{\Lambda}x\right)^2}{3\left(a^4+2\,a^2b^2+b^4\right)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{4\,\Lambda a^2b^2\cos\left(\sqrt{\Lambda}x\right)^2}{3\left(a^4+2\,a^2b^2+b^4\right)\sin\left(\sqrt{\Lambda}x\right)^2} \\
& - \frac{2\,\Lambda b^4\cos\left(\sqrt{\Lambda}x\right)^2}{3\left(a^4+2\,a^2b^2+b^4\right)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{a^4\frac{\partial}{\partial x}F\left(x\right)^2}{12\left(a^4+2\,a^2b^2+b^4\right)F\left(x\right)^2} - \frac{a^2b^2\frac{\partial}{\partial x}F\left(x\right)^2}{6\left(a^4+2\,a^2b^2+b^4\right)F\left(x\right)^2} - \frac{b^4\frac{\partial}{\partial x}F\left(x\right)^2}{12\left(a^4+2\,a^2b^2+b^4\right)F\left(x\right)^2} \\
& + \frac{2\,\Lambda a^4}{3\left(a^4+2\,a^2b^2+b^4\right)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{4\,\Lambda a^2b^2}{3\left(a^4+2\,a^2b^2+b^4\right)\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{2\,\Lambda b^4}{3\left(a^4+2\,a^2b^2+b^4\right)\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{\Lambda a^4}{12\left(a^4+2\,a^2b^2+b^4\right)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2} \\
& - \frac{\Lambda a^2b^2}{6\left(a^4+2\,a^2b^2+b^4\right)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2} - \frac{\Lambda b^4}{12\left(a^4+2\,a^2b^2+b^4\right)\cos\left(\sqrt{\Lambda}x\right)^2\sin\left(\sqrt{\Lambda}x\right)^2} + \frac{\Lambda^2a^2}{192\left(a^4+2\,a^2b^2+b^4\right)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^4}
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = & -\frac{\sqrt{\Lambda}a^4\sin\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}-\frac{2\,\sqrt{\Lambda}a^2b^2\sin\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}-\frac{\sqrt{\Lambda}b^4\sin\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}-\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{3\,\sqrt{\Lambda}a^4\sin\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}+\frac{3\,\sqrt{\Lambda}a^2b^2\sin\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}+\frac{3\,\sqrt{\Lambda}b^4\sin\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3} \\
& +\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& +\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{3i\,\Lambda^{\frac{3}{2}}a^3}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)}+\frac{3i\,\Lambda^{\frac{3}{2}}ab^2}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)} \\
& -\frac{\sqrt{\Lambda}a^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)}-\frac{\sqrt{\Lambda}a^2b^2\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)}-\frac{\sqrt{\Lambda}b^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)} \\
& -\frac{i\,\Lambda a^3\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{i\,\Lambda ab^2\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^2b^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{b^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^2b^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{b^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}+\frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2} \\
& +\frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}-\frac{3i\,\Lambda^{\frac{3}{2}}a^3}{8\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)^3}-\frac{3i\,\Lambda^{\frac{3}{2}}ab^2}{8\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)^3} \\
& +\frac{i\,\Lambda a^3\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}+\frac{i\,\Lambda ab^2\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}-\frac{\Lambda^2a^2}{16\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^4}
\end{aligned}
$$

$\Psi_1 = 0$

$$
\begin{aligned}
\Psi_2 = & -\frac{\sqrt{\Lambda}a^4\sin\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}-\frac{2\,\sqrt{\Lambda}a^2b^2\sin\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}-\frac{\sqrt{\Lambda}b^4\sin\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}+\frac{5\,a^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{12\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4} \\
& +\frac{5\,a^2b^2\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{6\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{5\,b^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{12\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{3\,\sqrt{\Lambda}a^4\sin\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}+\frac{3\,\sqrt{\Lambda}a^2b^2\sin\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}+\frac{3\,\sqrt{\Lambda}b^4\sin\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3} \\
& -\frac{5\,a^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{6\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{5\,a^2b^2\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{3\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{5\,b^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{6\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& +\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{\Lambda a^4}{12\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{\Lambda a^2b^2}{6\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& +\frac{\Lambda b^4}{12\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{i\,\Lambda^{\frac{3}{2}}a^3}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)}-\frac{i\,\Lambda^{\frac{3}{2}}ab^2}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)} \\
& -\frac{\sqrt{\Lambda}a^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)}-\frac{\sqrt{\Lambda}a^2b^2\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)}-\frac{\sqrt{\Lambda}b^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)} \\
& -\frac{i\,\Lambda a^3\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{i\,\Lambda ab^2\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{5\,a^4\frac{\partial}{\partial x}F(x)^2}{12\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{5\,a^2b^2\frac{\partial}{\partial x}F(x)^2}{6\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4} \\
& +\frac{5\,b^4\frac{\partial}{\partial x}F(x)^2}{12\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^2b^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{b^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{\Lambda a^4}{12\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}-\frac{\Lambda a^2b^2}{6\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2} \\
& -\frac{\Lambda b^4}{12\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}+\frac{i\,\Lambda^{\frac{3}{2}}a^3}{8\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)^3}+\frac{i\,\Lambda^{\frac{3}{2}}ab^2}{8\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)^3} \\
& +\frac{i\,\Lambda a^3\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}+\frac{i\,\Lambda ab^2\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}+\frac{\Lambda^2a^2}{48\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^4}
\end{aligned}
$$

$\Psi_3 = 0$

$$
\begin{aligned}
\Psi_4 = & -\frac{\sqrt{\Lambda}a^4\sin\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}-\frac{2\,\sqrt{\Lambda}a^2b^2\sin\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}-\frac{\sqrt{\Lambda}b^4\sin\left(\sqrt{\Lambda}x\right)^3\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}-\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{3\,\sqrt{\Lambda}a^4\sin\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}+\frac{3\,\sqrt{\Lambda}a^2b^2\sin\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3}+\frac{3\,\sqrt{\Lambda}b^4\sin\left(\sqrt{\Lambda}x\right)\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3} \\
& +\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{a^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& +\frac{a^2b^2\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{b^4\sin\left(\sqrt{\Lambda}x\right)^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{3i\,\Lambda^{\frac{3}{2}}a^3}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)}+\frac{3i\,\Lambda^{\frac{3}{2}}ab^2}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)} \\
& -\frac{\sqrt{\Lambda}a^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)}-\frac{\sqrt{\Lambda}a^2b^2\frac{\partial}{\partial x}F(x)}{(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)}-\frac{\sqrt{\Lambda}b^4\frac{\partial}{\partial x}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)} \\
& -\frac{i\,\Lambda a^3\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{i\,\Lambda ab^2\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^2b^2\frac{\partial}{\partial x}F(x)^2}{2\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{b^4\frac{\partial}{\partial x}F(x)^2}{4\,(a^4+2\,a^2b^2+b^4)F(x)^2\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}-\frac{a^2b^2\frac{\partial^2}{(\partial x)^2}F(x)}{2\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4} \\
& -\frac{b^4\frac{\partial^2}{(\partial x)^2}F(x)}{4\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4}+\frac{\Lambda a^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}+\frac{\Lambda a^2b^2}{2\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2} \\
& +\frac{\Lambda b^4}{4\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}-\frac{3i\,\Lambda^{\frac{3}{2}}a^3}{8\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)^3}-\frac{3i\,\Lambda^{\frac{3}{2}}ab^2}{8\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^3\sin\left(\sqrt{\Lambda}x\right)^3} \\
& +\frac{i\,\Lambda a^3\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}+\frac{i\,\Lambda ab^2\frac{\partial}{\partial x}F(x)}{8\,(a^4+2\,a^2b^2+b^4)F(x)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^2}-\frac{\Lambda^2a^2}{16\,(a^4+2\,a^2b^2+b^4)\cos\left(\sqrt{\Lambda}x\right)^4\sin\left(\sqrt{\Lambda}x\right)^4}
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"